Reverse Engineering Course

Dynamic Analysis

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Dynamic Analysis

- Easier to understand code functionality
  - Check values in registers, memory
  - Check stack: arguments and return addresses
- Virustotal

Dynamic Analysis

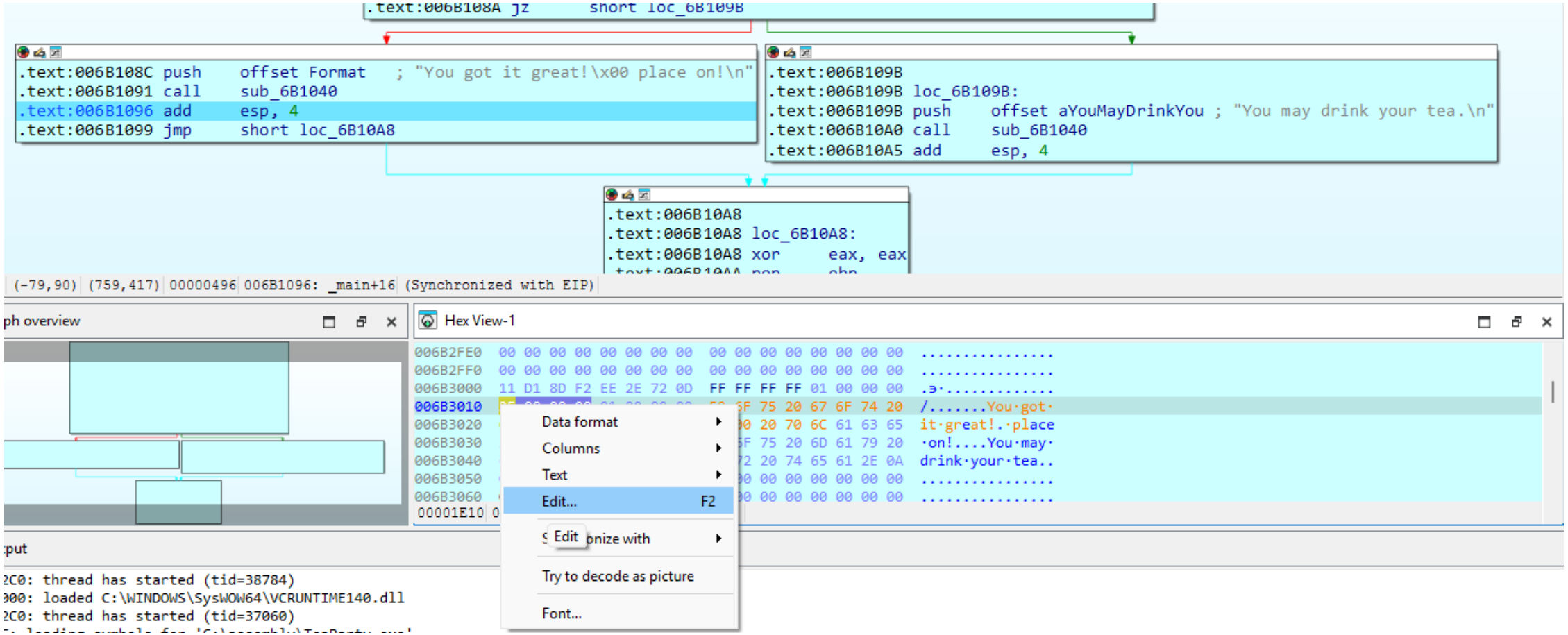
- Modify on the fly:
  - Registers
  - Code
  - Data
- Patch executable

TeaParty

- Modify registers to drink your tea
- https://data.cyber.org.il/reversing/TeaParty.zip



Tea Party – Edit Data



SimpleKey

- Crack SimpleKey.exe
- Jump to address – hex view – G
- https://data.cyber.org.il/reversing/SimpleKey.exe
- Challenge – edit your password on the fly

Change Code

- Edit -> Patch program -> Change byte
- Edit -> Patch program -> Assemble
- Use “Rename” memory address for jumps
- Apply changes to executable:
  - “Apply patches to input file”



IDA – Play with Patching

- https://data.cyber.org.il/reversing/BreakMe.exe
- Patch and apply patches to BreakMe.exe:
  - Step 1: Accept all but correct password
  - Step 2: Accept all passwords
  - Step 3: Don’t even ask for a password
    - Hint: Rename memory location





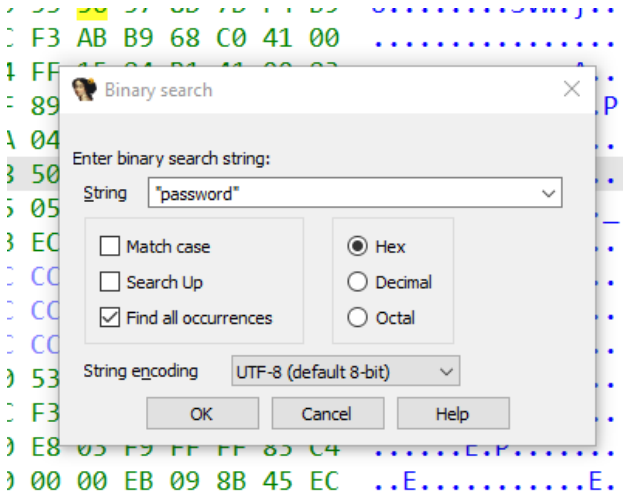
Finding main with the stack

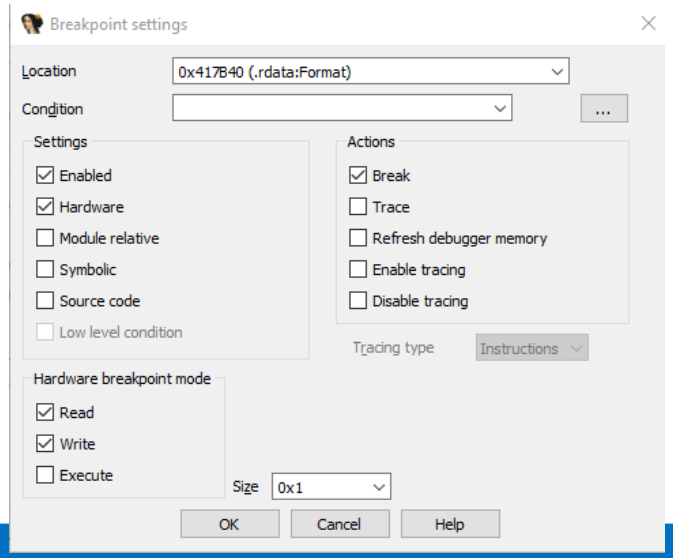
- Step 1: take whatever program prints and search
- Step 2: any print should be called originally from main
- Step 3: look at the stack- return addresses



Finding main with the stack

- IDA:
  - Hex view
  - Alt + B – find pattern
  - Debugger->Breakpoints->Add
    - Hardware
  - Run
  - View stack – return addresses
    - Ctrl+F7 – go up





GoUp

- https://data.cyber.org.il/reversing/GoUp.exe
- Solve as quickly as you can

Class Work – Patching Data

- Vegas
- Reach good message:
  - No code patching
  - No arguments

Class Work – Patching Code

- Patch InjectMe.exe
- Tip: study msdn WinMain
- https://data.cyber.org.il/reversing/TheNurse.rar

Hunt the flag

- https://data.cyber.org.il/reversing/ShabakFirst.exe



Class Work

- https://data.cyber.org.il/reversing/Secret.zip

Class Work

- Find the flag!
- https://data.cyber.org.il/reversing/FindTheFlag.pdf



Class Work

- https://data.cyber.org.il/reversing/MyDog5.exe
- Find the password